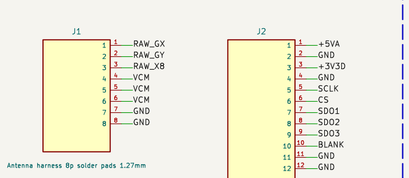
PF233-AB1A: AB1A receive analog board: 3 channels (GX, GY, X8), blanking, gain, 16-bit ADC

How to read: parts sit in functional blocks (dashed frames); every pin ends in a short wire with its NET NAME: equal names are connected (also between blocks). A cross = intentionally unconnected pin.
Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

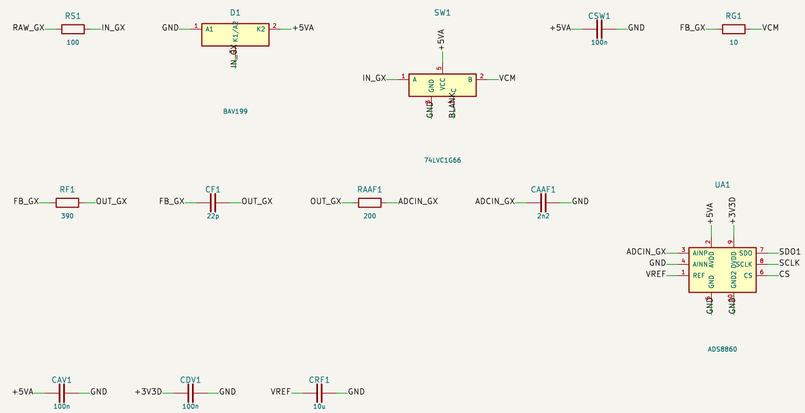
1 Connectors

J1: antenna harness (raw coil signals RAW_*, and VCM returns).
J2: main-board 2x6 header: +5VA, +3V3D, SPI (SCLK, CS, SDO*), BLANK.



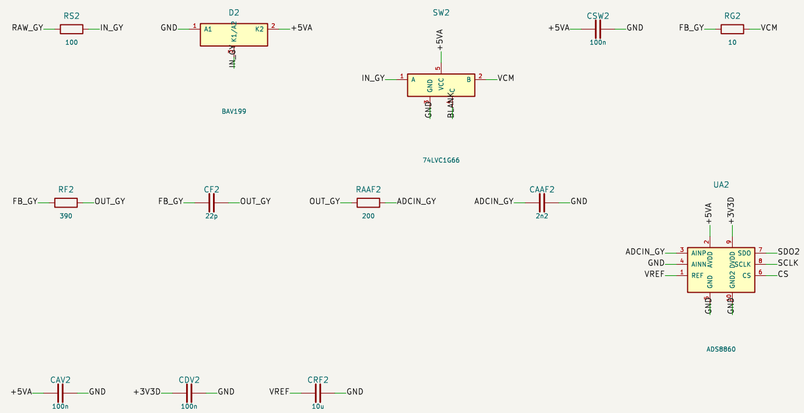
2 Channel GX

RAW_GX -> RS1 (series) -> D1 (BAV199 clamp to +5VA / GND) -> SW1 (74LVC1G66; BLANK opens the path) -> IN_GX -> op-amp U1 half A.
Non-inverting gain stage: RG1 to VCM, RF1 || CF1 feedback; OUT_GX -> RAAF1 / CAAF1 anti-alias -> ADCIN_GX -> UA1 (ADS8860, 16-bit SAR; REF = VREF with CRF1).
SPI: SCLK / CS from the main board, SDO1 back.



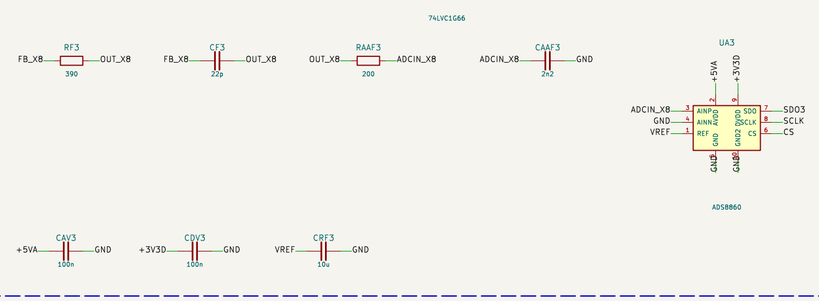
3 Channel GY

RAW_GY -> RS2 (series) -> D2 (BAV199 clamp to +5VA / GND) -> SW2 (74LVC1G66; BLANK opens the path) -> IN_GY -> op-amp U1 half B.
Non-inverting gain stage: RG2 to VCM, RF2 || CF2 feedback; OUT_GY -> RAAF2 / CAAF2 anti-alias -> ADCIN_GY -> UA2 (ADS8860, 16-bit SAR; REF = VREF with CRF2).
SPI: SCLK / CS from the main board, SDO2 back.



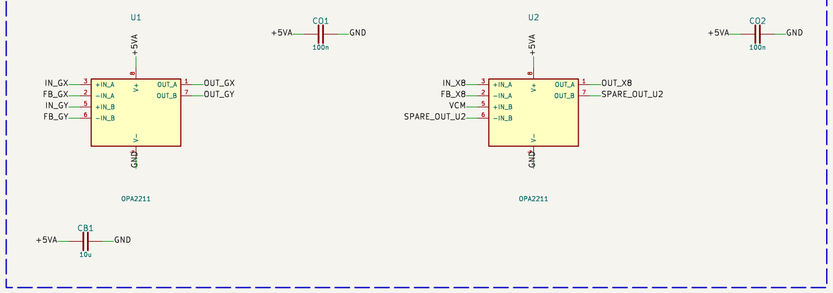
4 Channel X8

RAW_X8 -> RS3 (series) -> D3 (BAV199 clamp to +5VA / GND) -> SW3 (74LVC1G66; BLANK opens the path) -> IN_X8 -> op-amp U2 half A.
Non-inverting gain stage: RG3 to VCM, RF3 || CF3 feedback; OUT_X8 -> RAAF3 / CAAF3 anti-alias -> ADCIN_X8 -> UA3 (ADS8860, 16-bit SAR; REF = VREF with CRF3).
SPI: SCLK / CS from the main board, SDO3 back.



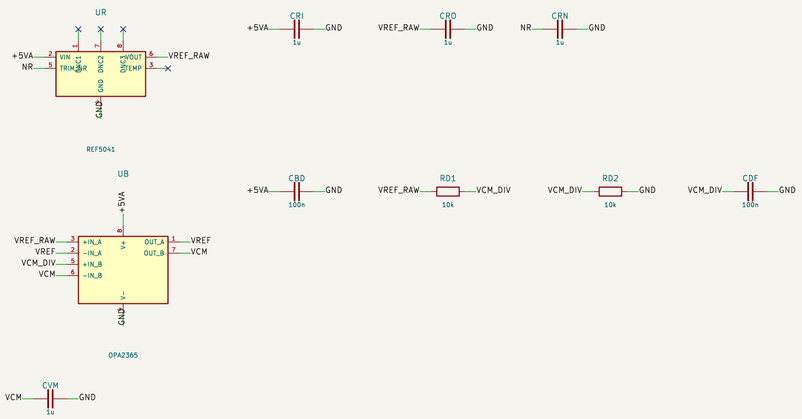
5 Gain-stage op-amps

OPA2211 dual op-amps (each half = one channel) from +5VA, decoupled CO*; CB1 bulk.



6 Reference and common-mode

UR (REF5041) -> VREF_RAW (CRI / CRO / CRN) -> UB (OPA2365) buffers VREF for the ADCs; RD1 / RD2 / CDF divide VREF_RAW to VCM_DIV -> UB other half -> VCM (CVM).



Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
AB1A receive analog board: 3 channels (GX, GY, X8), blanking, gain, 16-bit ADC
ARX / PHI-FOLD 233

Sheet: /		
File: PF233-AB1A_functional.kicad_sch		
Title: PF233-AB1A functional schematic		
Size: User	Date: 2026-10-26	**Rev: r3**
KiCad E.D.A. 9.0.9		Id: 1/1